Low-Power Digital Link Lab

An easy-language guide for your demo, report and instructor questions.

What problem are we solving?

A digital link sends bits through a noisy channel. More signal energy usually makes bits easier to recover. Error-correcting coding can achieve similar reliability with a weaker signal, but it adds bits. Our job is to find a workable balance between reliability, useful data rate, bandwidth, airtime and energy.

What did we build?

A browser demo that sends a typed message through a software model of a transmitter, channel and receiver. A separate Python study sends many known random bits, compares configurations, exports the results and supports one proposed design.

What is the target?

BER = 10^{-4}, or one wrong payload bit per 10,000 payload bits on average. BER is an error rate over a test population. It does not describe which bit will fail or guarantee a particular packet arrives.

What should I say in the first 30 seconds?

I tested how much signal energy different modulation and coding settings need for the same error rate. I can show a message entering the simulated transmitter, the signal being disturbed, the decoded output, and the full study that supports the recommended settings.

Where to look in this booklet

Pages 2-3 show the demo and signal chain. Pages 4-5 explain the experiments and final choice. Page 6 covers power and battery life. Pages 7-9 cover hardware and the viva.

Open the Digital Link Lab website

Show the demo in three minutes

Open the live experiment. Use a short message such as HELLO LAB. Keep the seed fixed when repeating a setup so that another person can reproduce it.

1. Start with an easy transmission

Click A. Clear signal. It runs QPSK, rate 1/2, at 10 dB. It still has modeled noise, but the signal is strong relative to the noise. Point to the original and recovered text.

2. Compare the noisy cases

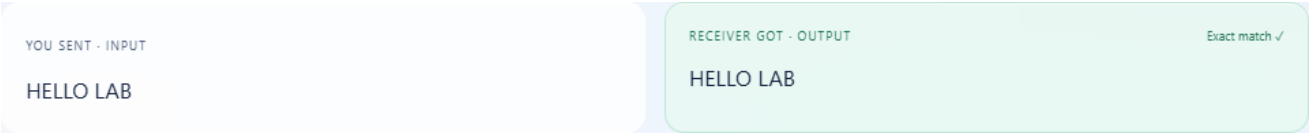
Click B. Noise, no protection and then C. Noise + protection. Both use QPSK at 2 dB. With HELLO LAB and seed 42, the separate 20,000-bit comparison counted 791 uncoded errors and 72 rate-1/2 errors. The message itself recovered correctly with protection. The BER comparison and typed message are separate tests.

3. Read the outputs

Recovered text is what the receiver inferred. Wrong bits divided by tested payload bits gives BER. A text error can change one character or make a UTF-8 sequence unreadable. A successful short message is a demo. Use the larger random-bit comparison and saved study to discuss reliability.

If both messages come through

Decrease Eb/N0 and run again, or choose a larger BER trial under Experiment options. If you see zero errors in a short run, report zero observed errors and name the number of tested bits.



Follow the signal through the link

Message > bits > add protection > shuffle > symbols > waveform > channel > recover symbols > undo shuffle > decode > compare bits.

Source representation and encoder

Text becomes UTF-8 bytes and then bits. The study generates bits directly. The convolutional encoder adds related bits using a short memory of previous inputs. At rate 1/2 it emits two coded bits for each input bit before packet overhead. Rates 2/3 and 3/4 omit selected coded bits through puncturing.

Interleaver and mapper

The interleaver reorders coded bits and the receiver reverses that order. This can spread groups of channel errors before decoding. BPSK uses two possible symbols and carries one coded bit per symbol. QPSK uses four symbols and carries two coded bits per symbol.

Pulse shaping and channel

Root-raised-cosine (RRC) shaping makes a sample-by-sample waveform and controls bandwidth. The receiver uses the matching filter. Path loss weakens the signal. Additive white Gaussian noise (AWGN) models random disturbances. The full BER sweep uses the ideal symbol-level equivalent to run faster.

Demodulator and soft Viterbi decoder

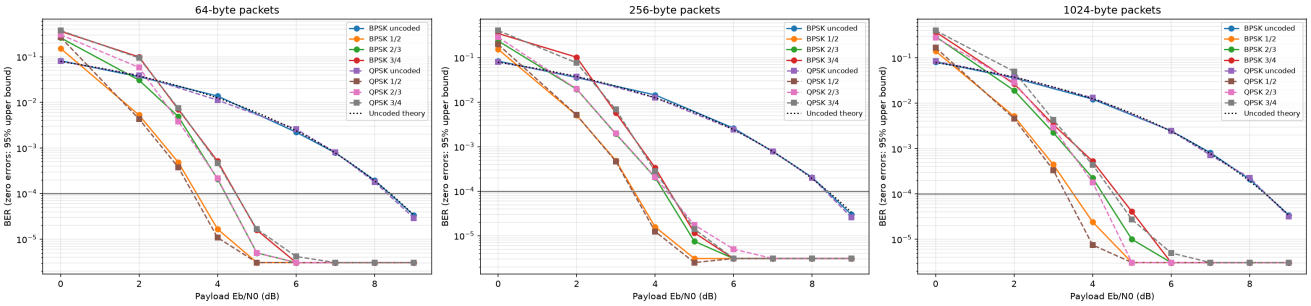
The receiver estimates which symbols were sent. Soft decisions keep a measure of how believable each possible bit is. Viterbi uses those scores and the encoder's rules to choose the most likely valid bit sequence. The simulator then compares recovered payload bits to the original.

What is assumed about the receiver?

The model already knows timing, carrier and channel gain. Physical radios would need to find the start of packets and correct timing and frequency errors.

Read the graphs and tables

In the BER plot, the horizontal axis is payload Eb/N0 in dB. The vertical axis is BER on a log scale. Down means fewer bit errors. Left means less energy per useful bit for the same target.



What is coding gain?

Read both curves at the same BER target. For the 256-byte QPSK screening curves, uncoded needed about 8.34 dB and rate 1/2 needed about 3.43 dB. Subtract to get about 4.91 dB. These are estimated curve crossings. The selected design uses a separately confirmed point at 3.75 dB.

What did the study vary?

The initial study used 204 simulation points and about 143.7 million payload bits. It compared BPSK/QPSK, uncoded and three code rates, and 64/256/1024-byte packets across Eb/N0. The results CSVs contain errors, bit counts and configuration comparisons.

Why is the data-rate chart flat?

Ideal AWGN BER depends on energy per bit relative to noise. At fixed Eb/N0, higher useful bit rate still increases the instantaneous RF power needed under the same link assumptions. The rate plot uses this model with 10, 50 and 100 kbit/s. It represents a calculation using the BER results.

Defend the final recommendation

Choose QPSK, convolutional rate 1/2, 256-byte payloads, 50 kbit/s active payload rate and 3.75 dB payload Eb/N0 under the stated simulation assumptions.

Why this combination?

Rate 1/2 adds strong protection. QPSK packs two coded bits into each symbol and makes this protected design more bandwidth-efficient than BPSK at the same useful bit rate. The selected point uses about 67.7 kHz of estimated bandwidth and 40.96 ms of payload airtime, within the study limits of 150 kHz and 50 ms.

How did we check the target?

A fresh simulation counted 39 errors in 1,228,800 payload bits. BER = 39 / 1,228,800 = 3.17 x 10^{-5}. The approximate one-sided 95% upper estimate across batches was 5.20 x 10^{-5}, below the 10^{-4} target. Batching helps account for bursts of decoded errors.

Why didn't we use the lower screening result?

A BPSK candidate at 3.50 dB returned 124 errors in 1,228,800 fresh bits. Its BER was about 1.01 x 10^{-4} and its upper estimate was about 1.35 x 10^{-4}. It failed the confirmation criterion. Small BPSK/QPSK differences in ideal AWGN can come from the random sample.

What does packet size change?

A fixed number of encoder tail bits takes a greater share of a short packet. A longer packet also takes longer to send and can contain an error even when BER is low. We compared 64, 256 and 1024 bytes. The chosen 256 bytes fit the modeled airtime limit.

Work through power and battery life

A link budget adds gains and losses to estimate how much RF power must leave the transmitter for enough energy to reach the receiver.

1. Estimate the required receiver input

For the worked example, use 50,000 useful bits/s, thermal noise density -174 dBm/Hz, a receiver noise figure of 6 dB, 3 dB margin and 3.75 dB target Eb/N0. The model gives -174 + 10 log10(50,000) + 6 + 3 + 3.75 = **-114.26 dBm**.

2. Add the path loss

For the example of 100 m free space at 915 MHz, calculated path loss is 71.68 dB and total antenna gain is assumed to be zero. The modeled transmit output is -114.26 + 71.68 = **-42.58 dBm**. These are the inputs and output of a worked model.

3. Calculate average electrical power

The traffic assumption is 1,000 useful bits/s and the active rate is 50,000, so the transmitter is active about 2% of the time. Active circuitry is modeled at 0.5 W and idle at 0.01 W, with a small additional RF amplifier term. Average = 0.02 x active power + 0.98 x idle power, about **19.8 mW**.

4. Convert usable battery energy to hours

10 Wh x 80% usable = 8 Wh. Runtime = 8 Wh / 0.0198 W = **about 404 hours**. This covers the modeled transmitter under the assumed traffic and power values. The result depends strongly on sleep current, actual activity, hardware and losses.

Two kinds of power

RF output in dBm describes the signal sent. Electrical power in watts describes what the circuit draws. A gain setting in radio software is a control value. A bench instrument is needed to relate that control value to real RF output.

What a bench test would prove

Stage 1 uses Python, NumPy, SciPy and Matplotlib for the study, and browser JavaScript for the demo. Convolutional coding fulfills the coding option in the brief. MATLAB and GNU Radio were not used for this software stage.

Proposed classroom cable setup

Laptop USB > bladeRF TX > RF attenuator + 50-ohm coax > RTL-SDR RX > USB back to laptop. USB moves digital samples. Coax moves the RF signal. No antenna is required for a conducted test. The lab must check the exact radio ports and input/output limits and select the attenuation before connection.

What still needs to be built and measured?

Add radio interfaces, packet framing, a preamble to acquire packets, synchronization, and a CRC to detect corrupted packets. Send known packets and log both payload BER and missing packets. Use the lab analyzer to relate gain settings to actual RF output, and measure electrical draw with the appropriate meter.

Where does a battery fit?

The laptop can run the first USB-powered bench demo. For a physical battery trial, use the exact radio's supported power arrangement, meter the transmitter's consumption, and log operating duty. Battery runtime from those measurements must say which equipment it covers.

What can I claim now?

The simulator meets its BER target at the recommended setting under the modeled channel. The conducted test would validate packet recovery and measured performance on a specific bench setup. Range, interference, obstacles and real battery life require additional measurements.

Instructor questions and answers

Use these as speaking notes. Explain an answer in your own words and point to the relevant demo, chart or calculation.

Why use coding when it increases bits?

Redundancy helps repair errors and lowers the required energy per useful bit. The costs are more coded symbols, processing, and potentially more bandwidth or airtime.

Is Eb/N0 the same as dBm?

Eb/N0 is an energy-to-noise ratio in dB. dBm is power relative to one milliwatt. The link budget uses the bit rate and noise assumptions to convert the reliability requirement into estimated RF power.

Does zero wrong bits prove the target?

It only means that run observed none. State the test size. Use repeat runs and uncertainty estimates to support a claim about the true error rate.

How do I know a packet was lost?

A real link needs known packet identifiers or a sequence count. A missing packet has no recovered bits for a direct bit comparison, so report missing packets as well as BER.

Why use both a waveform demo and a sweep?

The waveform demo makes the signal processing visible. The symbol-domain sweep tests many more bits efficiently under ideal matched filtering and synchronization.

Where is each deliverable?

The site shows message recovery, animation, charts and downloadable results. ber_measurements.csv has raw counts. configuration_summary.csv has coding gain. power_tradeoffs.csv has modeled power. final_confirmation_summary.csv has the fresh comparison. The submission PDF contains the recommendation and worked budget.

Could another configuration win?

Yes. A new throughput, airtime or bandwidth requirement can change the choice. Hardware measurements could also change the selected signal level and the battery estimate.

More viva questions

Practice these after you can explain the project in one minute.

Why use QPSK if its ideal BER is close to BPSK?

QPSK carries two coded bits per symbol, so it needs roughly half the symbol rate at the same coded bit rate. With comparable shaping and a coherent ideal receiver, both have the same theoretical uncoded BER at equal Eb/N0.

What do the waveform and constellation show?

The waveform plots sample amplitude over time. The constellation plots recovered symbols on I and Q axes. QPSK has four ideal positions and BPSK has two. Noise spreads the received samples around their ideal positions.

How can the decoder repair errors without the original?

It knows the encoder's rules and uses the reliability of received values to choose a likely bit sequence. The original payload is used afterward to score the result.

Does this interleaver remove noise?

It only changes the ordering of coded bits. After the receiver reverses that ordering, a cluster of errors can be spread out. An ideal independent-noise channel does not give interleaving the same advantage as a bursty channel.

What is the difference between data rate and duty cycle?

The 50 kbit/s rate applies while a packet is sent. The 1 kbit/s traffic rate is an assumed average over time. Their ratio gives 2% active time in the simple model. Preambles, retries and listening can change it.

What can I reproduce in front of the instructor?

Run the two 2 dB presets with the same message and seed. Explain the live bit counts. Then open BER & tradeoffs and show the saved results near the target, the selected point and the worked report.